INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 2

1. Experiment Title:

Updating and Deleting Rows

2. Aim:

Update values in selected rows, increase a quantity, delete a row, and verify the table after each operation.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

UPDATE modifies existing records. The WHERE clause identifies which row should change. DELETE removes records that match its WHERE condition. SELECT is used to verify the changes.

5. Algorithm / Procedure:

1. Create the stock table and insert the three setup rows.
2. Update item_code 1 so its quantity becomes 80; display the table.
3. Add 10 to the quantity of item_code 2; display the table.
4. Delete item_code 3; display the remaining rows.

6. Program / Queries / Commands:

```
CREATE TABLE stock (
    item_code INT,
    item_name VARCHAR(30),
    quantity INT
);

INSERT INTO stock (item_code, item_name, quantity)
VALUES
    (1, 'Notebook', 50),
    (2, 'Pen', 120),
    (3, 'Bag', 0);

-- Task 1
UPDATE stock
SET quantity = 80
WHERE item_code = 1;
SELECT * FROM stock ORDER BY item_code;

-- Task 2
UPDATE stock
SET quantity = quantity + 10
WHERE item_code = 2;
SELECT * FROM stock ORDER BY item_code;

-- Task 3
DELETE FROM stock
WHERE item_code = 3;
SELECT * FROM stock ORDER BY item_code;
```

7. Input:

Setup rows: (1, 'Notebook', 50), (2, 'Pen', 120), (3, 'Bag', 0).

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Stage	item_code	item_name	quantity
After Task 1	1	Notebook	80
	2	Pen	120
	3	Bag	0
After Task 2	1	Notebook	80
	2	Pen	130
	3	Bag	0
After Task 3	1	Notebook	80
	2	Pen	130

9. Result:

The SQL statements for Experiment 2 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is the use of UPDATE?
2. Why is WHERE important in UPDATE and DELETE?
3. What happens if DELETE is run without WHERE?
4. How can a value be increased using UPDATE?